



eCBHFA

Road safety module

Volunteer guide for empowering communities

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Volunteer guide

The International Federation of Red Cross and Red Crescent Societies (IFRC) is the world's largest volunteer-based humanitarian network, reaching 150 million people each year through our 191 member National Societies. Together, we act before, during and after disasters and health emergencies to meet the needs and improve the lives of vulnerable people. We do so with impartiality as to nationality, race, gender, religious beliefs, class and political opinions.

Our strength lies in our volunteer network, our community-based expertise and our independence and neutrality. We work to improve humanitarian standards, as partners in development and in response to disasters. We persuade decision-makers to act

at all times in the interests of vulnerable people. The result: we enable healthy and safe communities, reduce vulnerabilities, strengthen resilience and foster a culture of peace around the world.

The IFRC is proud to host the Global Road Safety Partnership (GRSP), which has been working collaboratively with National Societies, governments, civil society and the private sector to improve road safety and decrease the global road trauma burden. The Sustainable Development Goals contain two specific references to improving road safety, which is an important addition to the development agenda and one that the IFRC, through the GRSP, are striving to meet.

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Foreword

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This eCBHFA volunteer guide will equip you with the basics of road safety – how it works, and how to support your community with the information and motivation to protect their families. No two communities, or even families are the same, and so the community understanding and ability to adapt our work is critical in this effort.

You are part of a global movement to protect every person from preventable injury. Use of the road comes with unique challenges and represents one of the greatest risks we face in our daily lives. Using the information contained in this eCBHFA volunteer guide will assist in building knowledge among individual community members who can help to reduce the burden placed on every community by road traffic crashes.

Using the guides and toolkit

Road safety volunteer guide

This guide is designed for community volunteers who work directly with community members. The guide contains key information and discussion points to work with community members in understanding risks involved with using the road and how to minimise those risks.

The volunteer guide is comprised of seven topics which include knowledge, tools, talking points and activities for teaching communities in an interactive way.

Road safety toolkit

The toolkit is designed to support both the volunteer guide and the facilitator guide with interactive tools, games and illustrative material. The toolkit is designed to be adapted and modified to local contexts, taking advantage of both local expertise and knowledge and differences in vaccination schedule and approach. Materials have been specifically designed to be printed in black and white or full colour, and with low-resource environments in mind.

Road safety facilitator guide

This guide is designed for staff or trainers of trainers to conduct a training of trainers (ToT) or preparation course for staff and volunteers who will work directly with community members.

The facilitator guide should allow volunteers hands-on experience with all the toolkit materials. Diagnostic tools to better understand community perceptions, barriers and individual concerns are also included. It assumes a two-part training with a core community assessment exercise in between.

The community assessment will help volunteers to best prepare themselves for the unique opportunities and challenges in their communities. Central to the approach is the belief that local adaptation, knowledge and ownership is key to successful road safety training.

Road safety topics

Topic	Contents
1	Road safety behavioural risk factors
2	Seat-belts and child restraints
3	Speeding
4	Helmets
5	Drinking, drugs and driving
6	Distractions

Topic 1

Road safety behavioural risk factors

Overview

Road crashes cause death, injury or permanent disability to thousands of people every day and millions of people every year. Many things can contribute to a road crash and to the severity of the injury. The vehicle, the road, and people can all contribute in different ways to road crashes. The behaviour of road users may cause a road crash, or increase the level of injury, especially if they do one or more of the following unsafe behaviours:

- Fail to use a seat-belt or child restraints, or use them incorrectly
- Fail to wear helmets, or use them incorrectly
- Speed
- Drink alcohol or use drugs and drive
- Use the road when distracted

Road crashes also cause negative impacts to the mental health of the victims and their families, as well as to that of those who provide first aid or emergency care at the scene of a road crash.

Learning objectives

At the end of this session, learners will be better able to:

- Identify the major road safety risk factors including not using or misusing seat-belts and child restraints, not wearing or misusing helmets, speeding, unsafe pedestrian behaviour, drinking or taking drugs and driving and using the road when distracted
- Explain the threat of these risk factors to personal health

Tools

- ❑ Road safety behavioural risk factor

❑ Road safety risk assessment card

Road safety behavioural risk factors

❑ Road safety behavioural risk factor tool

Road traffic injuries are a leading cause of death and injury around the world. Each year approximately 1.35 million people die and millions more are injured or disabled as a result of road crashes. As there are more vehicles on the road, road traffic crashes are a fast-growing problem in every country¹.

Road safety is no accident. There are several behavioural risk factors that increase the chance of being in a road crash. Those risk factors are:

- Speeding
- Drinking or taking drugs and driving
- Getting distracted by mobile phone use or other distractions when using the road
- There are also some behavioural risk factors that increase the chance of being injured if you are in a road crash. Those risk factors are:
- Not using seat-belts or child restraints
- Not wearing a helmet

If a person employs one of those behavioural risk factors, then they are at a higher risk of being in a road crash. So, they will be at a higher risk of having a serious injury with permanent effects (such as a permanent disability) or even a fatal injury. Environmental factors such as the condition of the road, the level of enforcement of road traffic laws, the weather and the time of day, the availability of road facilities for cyclists and pedestrians (e.g. separated bike lanes, footpaths, etc.) may also influence road user behaviour and crash risk. However, these things are out of an individual's control.

The impact of being in a road crash is not only physical but it is also psychological. Being in a road crash can affect the psychological well-being of the victims suffering from a permanent disability and of family members and friends who have lost a loved one. Such an incident could put a great deal of emotional pressure on those affected and they might require psychosocial support and possible psychiatric intervention.

In addition to the physical and psychological effects, road crashes have a large impact on countries' economies. Road crashes are the number one killer of young people between 5 and 29 years of age. Losing people of such a young age leads to the loss of productivity, which in turn affects the overall economy.

Reflection

1. Name one behavioural risk factor of road safety.
2. What kinds of activities can keep us safer when we are on the road?

¹ WHO, Global status report on Road Safety, 2018.

Pedestrians

Most of us are sometimes pedestrians – people who walk from one place to another. Walking is a basic and common mode of transport in all societies around the world. Virtually every trip begins and ends with walking. Walking comprises the sole means of travel on some journeys, whether a long trip or a short stroll to a shop. In other journeys, a person may walk for one or more portions of the trip, for example, walking to and from bus stops, with a bus trip in between². More than 270,000 pedestrians are killed each year as a result of road incidents and accidents. Many leave their homes as they would on any given day never to return. Globally, pedestrians make up 23 per cent of road traffic deaths.

We know that walking is good for us because we are getting more physical activity that can help to prevent noncommunicable diseases, but unfortunately, in some situations, increased walking can lead to increased risk of road traffic crashes and injury. Pedestrians' vulnerability is further heightened in settings where traffic laws are inadequately enforced³.

So can we stay safe when we are pedestrians?

Most pedestrians are hit or killed when they are crossing the road or when walking or standing at the side of the road. Pedestrians are also hit or killed when they fail to use traffic lights properly and try to run across the street when it is not indicated as safe for pedestrians to cross.

More pedestrians are hit or killed at night, when they may not be as visible. More people are hit at twilight or just as the night is approaching at sunset and the first hour of darkness.

Each country differs on whether more pedestrians are hit or killed more often during the week or over the weekend.

People who walk and text or read their phone as they move are at a higher risk of being hit or killed because they are not paying attention to the traffic around them. When wearing headphones and walking, it is also very difficult to hear cars approaching or car horns, so pedestrians become distracted and are more vulnerable.

Children often like to play games in the street like football as it provides a lot of room for play. Advise children to play in parks and open spaces where vehicles are not permitted.

To stay safe, it is suggested that pedestrians:

- Take extra caution when crossing the road.
- Be sure to cross at lighted pedestrian crossings whenever possible when it indicates it is safe to cross.
- When walking across or along the road, walk as far from the edge of the road as possible.
- Wear bright clothing. If reflective strips or reflective clothing or lights are available to pin to your back and front of your clothing, this can save your life while you are on the road because you are now more visible, especially at night.

² WHO, Pedestrian safety: A road safety manual for decision-makers and practitioners, 2013.

³ Ibid.

- Do not text or look at social media on your phone while you walk along the road or while trying to cross the road. Keep your attention on the traffic around you.
- When wearing headphones, do not wear noise-cancelling headphones or keep your sound so high that you can no longer hear road traffic.
- Do not play in the street.

Reflection

1. What are the biggest risks for pedestrians from cars or motorized vehicles?
2. What are three ways to keep yourself safe from road accidents when you are a pedestrian?

Country road safety trends

Major risk factors differ from one context to another, from one country to another and from one area to another, depending on things such as the road network and our behaviour on the road. Road traffic data published by the relevant authorities (e.g. Ministry of Interior or the road safety lead agency) are a good source of road traffic data, although some countries do not record detailed road crash information. The analysis of such reports gives an idea of the major risks in your country. These may be one or more of the above-mentioned risk factors.

Reflection

- What are the behavioural risk factors that could lead to road crashes?
- What can you do to reduce the severity of injury if you are involved in a road crash?
- List some of the effects of road crashes mentioned during this session.
- Do you think road safety is an issue in your country?
- What can be done to improve road safety in your country?

Assessment

Road safety risk assessment card

In order to identify the risk of being in a crash, look at the “Road safety risk assessment” card. The assessment card has questions on whether community members use the road safely and the frequency of any risky behaviours. Participants can accumulate points for the safe behaviour they practise. On the other hand, points will be deducted if they practise unsafe behaviour. The resulting score would be the level of risk of them being in a crash or getting severely injured due to unsafe behaviour (high, medium or low).

Your own score does not have to be shared with the other volunteers but can be kept by you for personal reference. As you learn more about road safety, you might wish to improve your score.

Wrap-up

Road crashes cause many deaths and serious or permanent injuries every day. Our behaviour on the road can contribute to road crashes and our risk of injury. Risky behaviours include failing to use a seat-belt or child restraints, or using them incorrectly, failing to wear helmets, or using them incorrectly, speeding, drinking alcohol or using drugs when driving or using the road when distracted. Pedestrians in particular are very vulnerable. There are ways to keep yourself safer on the road, whether as a driver or rider or pedestrian, and you can help your community to practise safer behaviour.

Topic 2

Seat-belts and child restraints

Overview

People who use seat-belts and child restraints in a moving vehicle can help protect themselves from serious and fatal injuries to the head, chest and abdomen. This is applicable if you are sitting in the front or rear seat of the vehicle.

When it comes to choosing a child restraint, it is important to consider the child's age, weight and height. Different types of restraints are needed as the child grows.

Learning objectives

At the end of this session, learners will be better able to:

- Identify the types of injuries that a person can sustain if they do not use seat-belts or child restraints
- Identify the appropriate type of child restraint for every child age, weight and height

Tools

- Wearing versus not wearing a seat-belt
- How to wear a seat-belt
- Types of child restraints
- Safe in my vehicle chart
- Safe in my vehicle photos

Car capacity

In any vehicle in/on which you are a passenger, it is important to take note of the capacity of the vehicle and not have more people in the vehicle than there are functional seat-belts. For example, in a car that has four seat-belts, the total capacity of the car is four people – regardless of the age or size of the passenger. When you put more people in the car than there are seat-belts simply because there is “enough” room, you put all the people in the vehicle at risk because the unbelted person can injure the others in the case of a crash.

If the vehicle in which you are riding is meant to have four seat-belts, but only two seat-belts are operational, then the safe capacity of that vehicle is just two persons. On a motorcycle or other vehicle that is not designed with seat-belts, the safe capacity is the number indicated on the vehicle as the capacity limit. On a motorcycle, this number is typically no more than two people.

Seat-belts

Seat-belt and child restraint protection from injury

❑ Wearing versus not wearing a seat-belt tool

Take a look at this video that shows what happens when a person does not wear a seatbelt: <https://www.youtube.com/watch?v=d7iYZPp2zYY>. But riding in the backseat without a seatbelt is still dangerous. Here is a video of what happens when the person in the rear seat fails to wear a safety belt: https://www.youtube.com/watch?v=lghzH_UdxSk

When a crash occurs, three types of collisions happen:

1. The vehicle collides with another object (vehicle, signpost, tree, etc.) or a human or animal
2. The body of the occupant hits the inside of the car
3. The internal organs of the occupant hit the chest wall or the skeletal structure

The second and third types of collisions may cause severe injuries. The proper use of seat-belts and child-restraints can help minimize any injury caused by the second and third types of collisions.

What happens the moment a crash occurs? Basically, a person who is not restrained by a seat-belt or a child restraint will keep moving at the speed of the vehicle before the collision has happened. This will cause the occupant's body to be thrown towards the car's interior, if not ejected partially or totally from the car. Being ejected from the vehicle can cause serious injury and death.

The most serious injury that can be caused by a crash is a head injury. Head injuries may happen at inappropriate speeds combined with the person not wearing a seat-belt or child restraint. Although seat-belts and child restraints do not prevent crashes from occurring, they can help to reduce the severity of the injury to people in the vehicle. So, seat-belts and child restraints increase the chance of survival in case a crash happens.

How do seat-belts protect vehicle occupants?

There are many videos available online that show the effect of not wearing a seat belt. Choose a video with caution that it is from a reliable source and shows the result on not wearing a seatbelt. A sample video is: <https://youtu.be/9tPzALVSSvs>

Other sources for videos include WHO's site for media campaigns: https://www.who.int/violence_injury_prevention/videos/en/

Or Australia's Traffic Accident Commission (TAC) at <https://www.tac.vic.gov.au/road-safety>

Explain that:

- A seat-belt helps to restrict the forward movement of the vehicle occupant when a crash occurs, and it can prevent them from being ejected.
- A seat-belt distributes the force of the crash over the strongest parts of the body, which protects the sensitive body parts like the head, the chest and the abdomen.
- A seat-belt protects other occupants in the vehicle. A person without a seat-belt can be thrown around and hit others.
- If a vehicle has an air-bag, this does not replace the need to wear a seat-belt. The air-bag is designed to work together with a seat-belt to protect occupants.

How should a seat-belt fit?

□ How to wear a seat-belt tool.

There is a right and a wrong way to wear a seat-belt. For the best protection against injury, the seat-belt should be worn as such:

The shoulder seat-belt should go upwards towards the opposite shoulder, fitting snugly on the chest, and should lie on the shoulder between the neck and the edge of the shoulder.

The shoulder seat belt should not be worn in the following ways:

The shoulder seat-belt should not be touching or rubbing against the face and neck. It is advisable to adjust the height of the seat if possible in order to ensure the seat-belt fits the right way.

OR

The shoulder seat-belt should not hang on the shoulder. It is advisable to adjust the height of the seat if possible in order to ensure the seat-belt fits the right way.

As for pregnant women, the right way of wearing the seat belt is as such:

The shoulder belt should not go right across the "baby bump", but should go underneath it, so that in the case of a crash, the pressure is not on the unborn child.

How do child restraints protect children?

- Large physical force is applied to the body when a collision occurs. Many parents mistakenly believe that they would be able to hold a child in their arms to keep them safe. This is not true. The physical force exerted in a crash

would cause a child to be torn from an adult/parent's arms, regardless of how hard they try to hold on.

- It is also dangerous to put a seat-belt around an adult holding a child. The safest way for a child to travel in a vehicle is in a child restraint appropriate for the weight and height of the child.
- A child restraint limits the head and neck movement in the event of a crash and protects the child's abdomen.
- A child restraint protects the heart and lungs from injury since the rib cage is flexible and cannot protect these sensitive organs.
- A child restraint distributes the force of the crash over the strongest parts of the body.
- A child restraint also protects the child in non-crash cases, such as sudden stops or swerving manoeuvres or a door opening while the vehicle is moving.

Reflection

1. List the types of collisions that occur during a crash. Which ones cause the most serious injuries?
2. How do seat-belts and child restraints protect individuals from injury?

Car restraints

Types of child restraints

□ Types of child restraints tool

There are four types of child restraints depending on the weight and height of the child as shown in the tool. It is important to fit the restraint that is appropriate for your child's age, weight and height.

How to fit a child restraint in a vehicle

Here's a video to see the proper way to fit the child seat into a vehicle: <https://youtu.be/uKwnh1jUHmU>

It is important to fit the child restraint correctly in the vehicle, so parents are advised to carefully read the instructions on the user manual provided with the child restraint. There are general points that should be followed to secure a child restraint, such as:

- Place the child restraint in the rear seat instead of the front seat. The front seats are closer to the air bags, which are dangerous for the child if they open when a crash occurs.
- Place the child restraint in the middle rear seat if the middle rear seat has a three-point seat-belt. This is the safest place to place the child restraint because it is far from the car doors.
- Place the child restraint in the rear seat behind the passenger seat and not behind the driver seat. This way the child can be put in and taken out of the car from the pavement and not the roadside, unless the pavement from which you put the child in and take the child out of the car is usually the one

that is on the driver's side.

- The child restraint should be placed firmly on the seat with the seat-belt wrapped in a tight way and with no movement of the child restraint to the front or the sides.

Headrests in a closed vehicle

Headrests in a car or other vehicle are part of the safety equipment in the event of a crash. Headrests should not be removed as they help to protect the neck from more serious injury if there is a crash.

Take a look at the following video to see how the headrest protects the neck of the passenger in a crash: <https://www.cnet.com/roadshow/videos/smarter-driver-neck-restraints-and-headrests-explained/>

This video further shows how the lack of a headrest resulted in a lack of cushioning when the passenger's head snapped back during the crash.

Reflection

Where is the safest place to put a child restraint in a vehicle?

Why should headrests not be removed?

Where should the headrest be in relation to your head?

Wrap-up

Seat-belts and child restraints help to protect vehicle occupants from serious injury and from ejection from the vehicle. There are different types of child restraints that match the different physical development stages of a child. It is important to wear a seat-belt in both the front and the rear seats of a vehicle at all times and it is important to seat children in the back of a car at all times.

Topic 3

Speeding

Overview

Speed has a great effect on the probability of being in a crash and on the severity of any injuries incurred. The higher the speed, the higher the chance of you being in a crash, the more severe your injuries and the more likely that the crash is fatal. Speed also greatly affects the severity of the injuries if a pedestrian, a cyclist or a two- or three-wheel rider is involved in a crash. Speed worsens the outcome of a crash, especially if vehicle occupants are not protected by seat-belts or child restraints and if two- or three-wheel riders are not properly wearing a suitable helmet.

Learning objectives

At the end of this session, you will be better able to:

- Define “speeding”
- Describe the effect of speed on crash and injury
- Describe how speeding increases the risk of a crash
- Identify road users that are most vulnerable to speeding vehicles

Tools

- Stopping distance tool

How does speed increase the risk of a crash and serious injury or death?

Speeding is defined as driving at an excessive speed (driving above the legal speed limit) or at an inappropriate speed (driving too fast for the conditions, but within the legal speed limit). So, driving speed is a safety issue when it is above the legal speed limit or it is not appropriate for the specific conditions, including bad weather, low visibility, the presence of vulnerable road users (for example, pedestrians on the road), or when there is a large amount of traffic on the road.

Speeding increases the risk of a crash due to the following factors. When a driver is speeding, they cannot see and react to oncoming hazards in enough time to stop the vehicle and avoid them - there is less opportunity to take preventive action. So, it will be highly likely that they lose control of the vehicle when they are faced with a hazard. In addition, speed affects the distance that the vehicle will travel after the driver brakes and the time it takes for it to come to a complete stop.

Excessive acceleration is when a vehicle goes faster than all others on the road, either from a stopping position or while passing others. Excessive acceleration is not only dangerous because your vehicle is travelling at a significantly higher speed than those around you, but also because it makes it more difficult to gauge the necessary stopping distance. The faster your vehicle goes, the longer it will take to bring the vehicle to a stop when necessary.

It is also important to note that driving well below the speed limit and much slower than the traffic around you can be just as dangerous. Other drivers may not be able to compensate for your lower speed and hit your car if you are not moving at the same speed as those around you. For example, if you are driving slowly in a lane and a car comes up behind you and overestimates your speed, they may run into the back of your car.

Take a look at this video to see an example of how driving too slowly can be just as dangerous as driving too fast: <https://youtu.be/4oqfodY2Lz0>

Note that this video is from the US and guidance should be adjusted based on the side of the road in which you drive in your country.

Speed and stopping distance

□ Stopping distance tool

According to brake.org.uk, “stopping distances include the distance travelled while the driver notices a hazard and applies the brakes (thinking or reaction distance), and while the vehicle comes to a full stop from its initial speed (braking distance)”.

There is also a direct link between the speed at which a vehicle is moving and the effect it has on the safety of its occupants. Imagine the human body inside a vehicle hitting the interior of the vehicle at its same speed, or the body of a pedestrian colliding with a speeding car. The human body cannot tolerate such an impact if it is at a high speed. The damage to the body can be so severe that

it can cause death. Studies have shown that even a small increase in speed can have a large impact on the severity of injury and risk of death. For instance, a five per cent increase in average speed leads to approximately a 10 per cent increase in the risk of injury, and a 20 per cent increase in the risk of death⁴.

There are several ways that you can ensure you have enough stopping distance to avoid hitting the car in front if it stops suddenly.

One method is the “three second rule”. To use this method while driving, look at a stationary object in the near distance such as a sign or streetlight ahead along the road on which you are driving. When the car or vehicle in front of you passes that object, you should be able to count to three slowly in your head before your car passes that same object.

Another method is to look ahead at brake lights of cars several cars in front of you. By looking ahead for car brake lights two to three cars ahead of you, you can better determine when you should break even if the person immediately ahead of you has not braked yet.

In summary, speed is a factor that worsens the impact of crashes, making even the smallest mistake by a road user a serious or fatal one.

Reflection

1. What is speeding?
2. Why is speeding dangerous?
3. What is excessive acceleration and why is it dangerous?
4. How is driving much slower than the speed limit dangerous?

What makes people speed?

Why do you think people in your community speed? Who does not speed, and why?

Drivers speed when they have a reason to rush (e.g. they are running late). They also speed when they feel that there is a “reward” in them arriving earlier. Some people speed for fun and thrill-seeking. Speeding occurs frequently when people believe there is no chance of getting caught by police or receiving a penalty, or when drivers think that they are very skilled, and their skills can prevent a crash. Also, some jobs can put pressure on employees to meet unrealistic deadlines to deliver goods, which may result in them believing that they must speed to achieve that.

All of these are reasons that people commonly employ to justify why they speed, but speeding is a major risk factor for being involved in a deadly crash, and it puts the lives of the car occupants and other road users at risk.

Vulnerable road users

Vulnerable road users, also called unprotected road users, are road users who are:

⁴ www.who.int. (2004). Violence and Injury Prevention. [online] Available at: https://www.who.int/violence_injury_prevention/publications/road_traffic/world_report/speed_en.pdf [Accessed 6 May 2019].

- more exposed to injury or death in a crash because they are not well protected
- at a higher risk of being in a crash
- at a higher risk of having serious outcomes resulting from that injury (disability or death)

Vulnerable road users are:

- pedestrians, especially children
- the elderly, and people with disabilities
- motorcyclists, cyclists and users of other motorized and non-motorized two- and three-wheelers

These road users are even more vulnerable in low and middle-income countries due to poor road infrastructure, lack of law enforcement, lack of awareness of safe road use behaviours, and limited availability of emergency and hospital services.

In many parts of the world, the roads are risky for pedestrians because they are mixed traffic roads. This is particularly a problem when the area is a highly populated urban area because the risk of a pedestrian being hit by a vehicle is high. Often, vehicles approach pedestrian crossing areas at high speeds and fail to give right of way to pedestrians. Also, when safety measures such as pedestrian crossings and footpaths, etc. are not provided, pedestrians are at risk. Sometimes pedestrian behaviour itself increases this risk when they do not use the available pedestrian bridges or crossings that have been provided for their safety. Other vulnerable road users such as cyclists may increase their risk of accidents if they do not use cycle paths.

It is common for bicycles and non-motorized two- and three-wheelers to have to share the road with heavy vehicles (buses and trucks) in many places, which adds to the risk of these vulnerable groups. Many countries do not have roads that are safe for this mix of different kinds of vehicles and road users.

So, it is very important that speed limits are appropriately set and respected, particularly when the road is being shared with vulnerable road users. In some countries, the speed limit is set lower in certain zones, such as around hospitals and schools, to protect pedestrians and other road users. Research indicates that while most vulnerable (unprotected) road users survive if hit by a car travelling at 30 km/h, the majority are killed if hit by a car travelling at 50 km/h⁵. Therefore, it is important to consider the impact of speed on those who are most vulnerable when setting speed limits.

Reflection

1. What is “speeding”?
2. What makes you speed/not speed?
3. How does speeding affect the likelihood of getting in a crash?
4. List the vulnerable road users and give an example of when each of them can be victims of speeding.

⁵ Speed management: a road safety manual for decision-makers and practitioners. Geneva, Global Road Safety Partnership, 2008, p.4.

Linking speed with other risk factors

As mentioned before, speed is a factor that worsens the impact of every crash, even if speeding was not the cause of the crash. Here is some information about how speeding is linked to the other behavioural risk factors.

- ***Speeding and the use of seat belts and child restraints***

It is impossible for passengers in a speeding car to keep hold of an unrestrained child. This is because the weight of the passenger increases in a crash. For example, a 5kg baby being held by a passenger will suddenly feel like 100kg, which is a 20 times increase in weight⁶. If the vehicle is moving at a high speed, whoever is holding the baby will not be able to keep doing so and the risk of injury for the baby becomes high. It is the same for an adult who is not wearing a seat-belt. The impact of the crash might cause them to leave their seat and hit the interior of the car or be ejected from the car. Seat-belts and child restraints are therefore an important safety measure to help protect against the impact of speed on the human body when a crash occurs.

- ***Speeding and the use of helmets***

Helmets help to protect the head and the brain from serious injury. However, the higher the speed at which a crash occurs, the more risk there is that the helmet would be damaged and injury to the head results. When helmets are not used, there is no protective cover for the head and any head or neck injury could be fatal. Sometimes helmets are used incorrectly (not buckling the straps or reusing an old or damaged helmet), so higher speeds may also lead to a higher risk of severe head injuries.

- ***Speeding and unsafe pedestrian behaviour***

As mentioned before, speeding means that a driver has less time to take preventive action when facing a hazard on the road, such as a pedestrian stepping out in front of their car, and they must make a sudden manoeuvre to avoid it. In that case, it will be highly probable that they lose control of the vehicle. Also, travel speed affects the distance that the vehicle will travel after the driver sees and reacts to the hazard and applies the brakes, and the time it takes for it to come to a complete stop. So, what if the hazard is a child running out into the middle of a road? If that child is 13 metres in front of the car, and the car is travelling at 50 km/h, the distance covered in the driver's reaction time (9 metres) and the stopping distance (23 metres) is more than the distance to the child. Consequently, the child will be hit by the car travelling at 50 km/h and the chances of the child surviving are small. If the car is travelling at 30 km/h, then the car can just be stopped before it hits the child.

Stopping distance at stop lights

Just as you should allow enough space between your car and the cars in front of you while driving, you can also prevent hitting cars in front of you in the event that you are struck from behind while waiting for a stop light or in heavy traffic. When you come to a stop in traffic, leave a car length of space between your car and the vehicle in front of you. When you do so and if a car hits you from behind, your car will not then hit the car in front of you and put you at greater risk of being crushed in your vehicle.

⁶ Speed management: a road safety manual for decision-makers and practitioners. Geneva, Global Road Safety Partnership, 2008, p.4.

Wrap-up

Travel speed has a direct impact on your risk of being involved in a road crash and on the severity of the crash outcome (death/injury). Speeding can lead to drivers losing control of their vehicle and it also makes it more difficult to protect vehicle occupants, pedestrians, and two- and three-wheel riders from severe injury or death.

Topic 4

Helmets

Overview

Helmets help to protect motorcyclists and cyclists from serious and fatal head injuries. These injuries can either be open or closed, with the closed ones being more dangerous. The type of helmet used and the way it is used are important safety factors. Helmets should be of good quality, should meet international quality standards and the chin straps should be buckled for maximum protection.

Learning objectives

At the end of this session, learners will be better able to:

- Identify how the head and brain can get injured during a crash on a motorcycle or bicycle
- Identify the level of protection that each type of helmet can offer
- Identify the correct way of wearing a helmet

Tools

- ❑ Head and brain structure
- ❑ Helmet components
- ❑ Types of helmets

Helmets against head injuries

- ❑ Head and brain structure tool

Head injuries

When users of motorcycles and bicycles are involved in a crash, they are usually at a high risk of having head and neck injuries. These injuries are the main cause of death, severe injury and disability among these road users. In addition to the physical consequences of head injuries, there is a social cost to those injuries. Survivors, their families and communities are the ones who pay this cost due to the specialized and long-term care that the survivors often need after a crash.

In order to understand how head injuries happen during a crash, take a look at the sketch of the head and brain in the toolkit.

The tool about the head and brain structure explains what is happening within the skull.

- The brain is engulfed and protected by a rigid skull.
- The brain rests on bones that make up the base of the skull.
- The spinal cord passes through a hole in the underside of the brain.
- Under the skull, adhering to the bones, is a tough tissue called the dura mater that surrounds the brain.
- Between the brain and the dura mater is a space containing cerebrospinal fluid that protects the brain tissue from mechanical shock.
- The brain floats in the cerebrospinal fluid but it can only move about one millimetre in any direction.
- The skull is covered by the scalp, which provides some additional protection.
- During a motorcycle or bicycle crash there are two principal mechanisms of injury to the brain: through direct contact and through acceleration–deceleration. Each mechanism causes different types of injuries.

When a motorcycle or bicycle is involved in a crash, the rider is usually thrown from the vehicle. If during this incident the rider's head hits an object, such as the ground or a pole or anything else, the head's forward motion is stopped, but the brain continues to move forward until it hits the inside of the skull. It then rebounds and hits the opposite side of the skull. This type of injury can result in anything from a minor head injury such as a concussion, to a fatal head injury. Head injuries that result from either contact or acceleration–deceleration injuries are themselves divided into two categories: open or closed head injuries.

Open head injury: involves a fracture of the skull caused by a hit or by penetration of an object.

Closed head injury: no fracture of the skull, but the brain can get bruised or swollen and the brain's tissues, blood vessels or nerves can get torn.

In order to help protect the head and brain from a serious injury, motorcycle riders must wear a properly fastened good quality helmet. Helmets provide an additional layer of protection for the head and thus protect the wearer from some of the more severe forms of traumatic brain injury.

How do helmets work

□ [Helmet components tool](#)

Helmets reduce the risk of serious head and brain injuries by lowering the impact of a force or collision to the head.

The World Health Organization, in its Helmets manual published in 2006, describes how a helmet works in three ways. The helmet:

- Reduces the force by which the brain hits the skull when the crash occurs. The soft material inside the helmet absorbs some of the impact and therefore the head comes to a halt more slowly.
- Spreads the force of the impact over a greater surface area so that it is not concentrated on specific areas of the skull.
- Prevents the head from touching the object because it acts as a barrier between the head and the object with which it is colliding.

Each part of the helmet plays a part in the protection of the head of the motorcyclist or cyclist from serious injury.

The shell

The shell is the strong outer surface of the helmet. This surface distributes the impact of a collision over a large surface area, which lessens the force before it reaches the head. It helps to provide protection against penetration by small, sharp and high-speed objects and it can also protect the padding inside the helmet from scrapes and knocks during daily use. Its exterior is hard and has a smooth finish.

Even though the shell is hard, it still can bend or break when hit by something hard. When the shell is damaged, it loses its protective properties and it will not be effective in protecting the head from injury if a crash occurs again. Therefore, it is extremely important to replace a damaged helmet.

The impact-absorbing liner

This layer beneath the shell is made of a soft, crushable padded material – usually expanded polystyrene, commonly called styrofoam. This dense layer acts like a cushion that absorbs the shock when the helmet stops, and the head continues moving.

The comfort padding

This is a soft layer made of foam and cloth that is next to the head. It helps keep the head comfortable and makes the helmet fit tightly.

The retention system/chin strap

The chin strap keeps the helmet on the head in a crash. One strap is connected to each side of the shell. Chin straps must always be buckled, otherwise the helmet will fly off the motorcyclist's or cyclist's head when a crash occurs. A fallen helmet will offer no protection against head injuries and its use would be pointless.

Any missing or non-functional part puts the motorcyclist or cyclist at a greater risk of a serious or fatal injury.

In summary, the helmet used should:

- be made of Expanded Poly-Styrene (EPS), Acrylonitrile Butadiene Styrene (ABS), Poly Carbon (PC) and Poly Propylene (PP)
- be replaced after three to five years of use
- be discarded after a crash
- not compromise the vision or hearing if it is a full-face helmet
- have a crushable liner between 1.5 cm and 3.0 cm thickness

Types of helmets

Types of helmets tool



Bicycle

- Decrease the risk of head and brain injuries. Helmets are very effective in decreasing this risk.
- Decreases the risk of a head injury by 69% and the risk of severe brain injury by 79%
- Effective in crashes involving motor vehicles as well as non-motor vehicles
- Prevent 2/3 of potential injuries to the middle and upper portions of the face or above the upper lip.



Full-face

- Protects the face, skull, chin and jaw
- Protects against impact
- Provides a maximum range of sight vertically and peripherally



Open-face

- Protects the skull
- Protects against impact
- Limited protection for the chin and jaw
- May or may not have retractable visors to protect the eyes

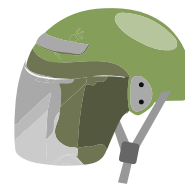


Half-head

- Protects the skull
- No protection for the chin and jaw
- Rarely equipped with visors
- May or may not have ear flaps connected to the retention system

Tropical

- Designed for very hot and humid areas (e.g. South and South-east Asian countries)
- Has ventilation holes to reduce the heat
- Light in weight



Note that adult helmets do not fit children's heads, so it is important that they use ones specific for children.

Other protection for motorcyclists

Helmets are the number one required safety equipment when riding on a motorcycle as they protect precious brains. Additional equipment that protects a motorcyclist in a crash includes:

- thicker clothing
- clothes that cover all exposed skin on arms and legs
- gloves
- knee pads
- elbow pads
- special jackets and pants to reduce number of injuries in a crash

Take a look at the following videos to see the cost to a motorcyclist to wear additional protective clothing as well as the cost to their body when they are not unprotected. <https://www.youtube.com/watch?v=mfks8yBkLzM>

Which motorcyclist is likely to have fewer injuries on the day of this accident as well as for years to come?

The capacity of each motorcycle indicates the SAFE number of people that a motorcycle can carry. Just because there is extra room does not mean it is safe – in fact, when you add additional people or cargo, you put everyone on the motorcycle at greater risk as it is more difficult to manage the load and balance of the vehicle, especially if there is an obstacle in the road or an unexpected stop by the vehicle in front of the motorcycle.

Reflection

- How does a helmet protect the motorcyclist or a cyclist in a crash?
- What properties should you look for in a helmet?
- What is the correct way to wear a helmet?
- What other types of protection can motorcyclists wear?
- How does additional clothing or equipment protect a motorcyclist?
- How many people can safely ride on a motorcycle?

Wrap-up

Helmets help to protect the head from serious and fatal injuries during a crash. Helmet wearing is effective when two factors are in place: the helmets are of good quality and the helmets are worn correctly (buckled up).

Topic 5

Drinking, drugs and driving

Overview

Due to the effect of alcohol on bodily functions, drink-drivers have a significantly higher risk of being involved in a road crash than drivers who have not consumed alcohol. Drivers using recreational drugs with or without alcohol put themselves, their passengers and pedestrians at greater risk.

Learning objectives

By the end of this session, learners will be better able to:

- Identify the legal BAC limit in their country
- Identify the alternative actions to driving they can take to keep themselves and others safe from a crash

Tools

- How much is too much tool
- Drug driving tool

What is alcohol, what are drugs and why do people consume them if they will be driving?

Alcohol is a liquid produced from the action of yeast on sugar. Examples of drinks that are produced in this way are beer, wine, arak, whiskey, vodka and gin.

Alcohol has many functions in society and bears important cultural, religious and symbolic meanings in most countries. However, we must be careful with its consumption because it is also a drug with many toxic effects. It is estimated that the number of deaths due to alcohol is roughly twenty-five times the number attributed to all illegal drugs combined⁷.

How does alcohol contribute to road crashes?

There are many videos available of the effects of a road crash. Several examples from the WHO page are: https://www.who.int/violence_injury_prevention/videos/australia_drink_driving_joey/en/ and <https://youtu.be/u6dMfkFkZHo>

The immediate effects of alcohol on the brain are either depressing or stimulating in nature, depending on the quantity consumed (see Table 1). In both cases, alcohol results in impairment which increases the likelihood of a crash because it results in:

- poor judgement
- increased reaction time
- lower vigilance
- decreased visual sharpness
- lower blood pressure
- reduced consciousness and respiration

Alcohol is also pain-relieving and causes a general insensitivity to pain.

So, knowing these debilitating effects of alcohol on the body, it is no wonder that it can increase the risk of a crash, even at relatively low Blood Alcohol Concentration (BAC) levels. However, as the BAC levels increase, effects on human performance become progressively worse. In addition to poor judgement and delayed reaction time, vision also deteriorates as more alcohol is consumed.

Not only does alcohol have a direct impact on crash outcomes, it can also affect the driver's decision to wear a seat-belt, use a helmet and speed.

The effects of alcohol can also negatively affect the diagnosis, management and treatment of injuries.

There are also a number of factors that affect the level of alcohol in the blood. Table 1 summarizes these factors. Many of the listed factors are not fixed, such as body weight, medication, tolerance, food intake, etc. So, it is hard to assess the level of alcohol in your blood yourself. Therefore, the safest thing to do is not to drive or ride when you consume any alcoholic beverage.

Table 1. Effect of alcohol on body and performance according to increasing BAC levels

The amount of alcohol consumed	The more alcohol is consumed, the higher the alcohol concentration in the blood.
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⁷ Global status report on alcohol and health. Geneva, World Health Organization, 2011.

The rate at which alcohol is consumed	The liver, the organ in our bodies that is responsible for processing alcohol, takes an hour to process the alcohol present in one standard drink. So, if more than one standard drink is being consumed in an hour, the excess alcohol will remain in the blood.
The amount of food in the stomach	Having food in the stomach before consuming an alcoholic drink delays the alcohol from reaching the small intestine. This means that alcohol will be delayed to reach the bloodstream as well.
The type of alcoholic drink consumed	Some types of alcohol interact with the small intestine in a way that makes it either absorb it fast or absorb it slowly.
Gender	When men and women consume the same amount of the same alcoholic drink, the women will usually have a higher concentration of alcohol in the blood. This is because biologically, women have less of the enzyme that breaks down the alcohol in the stomach.
Body weight and body type	Larger bodies have larger amounts of water. So, people with larger bodies who consume alcohol will have the alcohol less concentrated in their blood. Also, fatty tissue does not absorb alcohol. So, a person with more fatty tissue in their bodies, whether it is a male or a female, will absorb less alcohol and will have a lower blood alcohol concentration.
Liver health	A healthy liver will break down alcohol at the correct pace. An unhealthy liver will have an impaired function and that will affect the breakdown of alcohol.
Medication	Some types of medicines will slow down the breakdown of alcohol and will make it stay in the blood for longer periods.
Genetics	Your ability to break down alcohol can be genetic. For example, people from Asian descent have difficulty breaking down alcohol because of the different levels of activity of the related enzymes.
Tolerance	People who have been using alcohol for a long period of time or are heavy drinkers have a higher tolerance to alcohol. They will be able to break down alcohol faster and the body becomes less sensitive to alcohol.

What is a standard drink?

A standard drink is considered as 10g of pure alcohol in a drink. It is harmful for one's health to have more than one standard drink (SD) serving size per hour and it will increase the risk of a crash for the driver if he or she had more than one SD serving size per hour

The best way to avoid a drink-driving related crash is to have no alcohol at all before driving. The safest BAC is always zero grams of alcohol in the blood. If the driver consumes alcohol, then they should consider using alternative and safe methods to get home, such as a taxi or public transportation (if it is safe). It is

also recommended to plan ahead and appoint a designated driver who will not consume any alcohol or arrange a stay at a friend's place or at a hotel. Getting on the roads with alcohol in your system would risk your life and that of others.

The second-best way is to limit your intake of alcohol to the limits set by your government, bearing in mind the factors that affect BAC.

How is alcohol in the body measured?

□ How much is too much too?

The level of alcohol in the body can be assessed by measuring its amount in the blood and the measurement is called the Blood Alcohol Concentration (BAC). The amount of alcohol that is contained within the bloodstream can be measured by testing a small sample of blood or urine, or through analysis of exhaled breath. BAC is usually measured as:

- grams of alcohol per 100 millilitres of blood (g/100 ml)
- milligrams of alcohol per 100 millilitres of blood (mg/100 ml)
- grams of alcohol per decilitre (g/dl)
- milligrams of alcohol per decilitre (mg/dl)
- or other appropriate measure

In order to control the amount of alcohol that drivers consume, countries should set a "legal BAC limit" in their traffic laws. Different countries and even different states within the same country have different legal BAC limits that drivers can have. As a volunteer, check what the BAC limit is in your country/state and inform the community members about it. You can refer to the World Health Organization's Global Status Report on Road Safety 2018 that shows the BAC limit per country.

Legal BAC levels for driving vary from country to country, or state to state, throughout the world –ranging from 0g/100ml (zero tolerance to driving under the influence of alcohol) to 0.10 g/100 ml.

Drink-driving laws should be based on the scientific evidence that alcohol in the blood can negatively affect our ability to drive safely. It is very important to obey the drink-driving laws in your country because you can save your life and that of others.

Reflection

1. Name at least three effects of alcohol on the body.
2. Name at least three factors that can increase or decrease the effects of alcohol on the body.
3. What is the safest alcohol level in the blood?
4. What would you do to get home safely after drinking alcohol?

Drugged driving

❏ Drug Driving Tool

Use of recreational drugs or misuse of medication can make driving a car unsafe—just like driving after drinking alcohol. Drugged driving puts the driver, passengers, and others who share the road at risk.

Is drugged driving dangerous?

The effects of specific drugs differ depending on how they work in the brain. Marijuana or hashish can slow reaction time, cause the driver to weave on the road, impair judgment of time and distance and decrease coordination. Other drugs can make drivers more aggressive and reckless when driving while yet other drugs or medications can cause dizziness and drowsiness. All of these factors can lead to vehicle crashes.

Mixing alcohol with drugs make drivers even more impaired and dangerous. Even small amounts of some drugs can have a measurable and disastrous effect on a person's driving.

Why is drugged driving a problem in teens and young adults?

Teens and young adults are most likely to be influenced by peer pressure. The culture of taking recreational drugs to enhance experiences and enjoyment has increased enormously. Teen drivers are less experienced and are more likely than older drivers to underestimate or not recognize dangerous situations. They are also more likely to speed and allow less distance between vehicles. When lack of driving experience is combined with drug use, the results can be tragic. Car crashes are the leading cause of death among young people aged 16 to 19 years in many countries.

What steps can people take to prevent drugged driving?

Because drugged driving puts people at a higher risk for crashes, and if you are planning to use drugs and alcohol, come up with a plan to avoid getting behind the wheel of a car while impaired. Steps people can take include:

- offering to be a designated driver
- appointing a designated driver to take all car keys
- getting a ride to and from parties where there are drugs and alcohol
- discussing the risks of drugged driving with friends in advance

Reflection

1. Name at least three effects of drugs on the body.
2. What can happen if you mix drugs and alcohol?
3. What is the safest amount of drugs you can take before driving?

4. What would you do to get home safely after using drugs?

Wrap-up

Drivers who consume alcohol or use drugs increase their risk of a road crash. The safest blood alcohol concentration is zero and the safest amount of drugs in the driver's body is none. It is important to plan ahead for alternative means of transport in case a driver has consumed alcohol or is using drugs.

Topic 6

Distractions

Overview

Road crashes are often caused by distractions to the driver or pedestrian. These may include the driver using technology like mobile phones, GPS devices and tablets, but may also be passengers in the vehicle who are distracting the driver. Pedestrians may be distracted so that they forget road safety and become a hazard to drivers. This topic explores distractions and how to reduce road crashes due to distractions.

Learning objectives

At the end of this session, learners will be better able to:

- Identify the different types of distraction that can happen while driving or being a pedestrian
- List the effects of a distraction on driving behaviour
- Identify safe behaviour against mobile phone use while driving or being a pedestrian

Tools

- Types of driver distraction
- Distraction game

What is distraction?

- Types of driver distractions tool

Distraction is defined as “the diversion of attention away from activities critical for safe driving towards a competing activity”⁸.

⁸ Regan, Michael & A Lee, M & D. Young, J. (2008). Driver Distraction: Theory, Effects and Mitigation.

When drivers are behind the wheel, it is their “primary task” to be doing this activity. When they start doing other things that are not related to driving or “secondary tasks”, their attention will be temporarily divided between the two tasks. If that “secondary task” is a phone call, then the driver is using his/her cognitive/thinking resources to analyse two things: the driving situation and the phone conversation. This leads to the driver’s reduced attention to the driving task, impaired decision-making and driving performance.

There are four types of driver distraction:

1. Visual: looking away from the road for a task not related to driving
2. Cognitive: thinking about a subject of conversation instead of focusing on and analysing the road situation
3. Physical: when the driver holds or operates a device rather than steering with both hands, e.g. dialling a phone or leaning over to tune a radio that may lead to rotating the steering wheel
4. Auditory: responding to a ringing mobile phone, or not responding to sounds such as an ambulance siren if a device is turned up so loud that it masks other sounds

Note that sometimes more than one of these types of distraction may occur at the same time.

Sources of driver distraction

Sources of driver distraction may be internal or external.

- Internal: comes from inside the vehicle – for example activities like talking to other passengers, eating, drinking, smoking, grooming, adjusting the radio or air-conditioning and using technology (phone, GPS, CDs/DVDs, or other portable devices)
- External: comes from outside the vehicle. Such sources can be activities like looking at roadside advertising, other vehicles, shop windows, etc.

The danger that each source of distraction poses on road safety depends on the effect of the source of distraction on driving behaviour, as well as the frequency and duration of the behaviour.

Mobile phone use

Mobile phone use receives the most attention in this topic because it is the source of distraction that is currently receiving the biggest attention from researchers, the police, and other road safety stakeholders. This is because mobile phone use greatly contributes to road crashes, and also because phone use and their functions are increasing. For instance, phones were previously only used for texting and for making phone calls. More recently phones are also used to check email, social media accounts, play games, schedule events and watch videos.

Also, new technology like Bluetooth is encouraging the use of mobile phones by drivers in hands-free mode and via voice-activation.

The effect of mobile phone use on road crashes

Using mobile phones is considered as all four types of distraction together:

1. Visual distraction: takes the driver's eyes off the road
2. Cognitive distraction: the driver's mind is not focused on the road
3. Physical distraction: the driver's hands are off the steering wheel
4. Auditory distraction: the driver responds to the ringing phone. However, the duration of this distraction is shorter than the other types of distraction⁹.

This distraction can impair driving behaviour in the following ways:

- Causes a longer reaction time to detect and respond to an unexpected event or hazard.
- Makes it hard to maintain correct lane position.
- Causes slower reaction / braking, and when braking occurs, it is usually more intensive with shorter stopping distances.
- The driver will drive more slowly or at an inappropriate speed.
- Slows reaction to traffic signals or does not see them at all.
- Reduces the field of view since the driver is more likely to look straight ahead and not at the sides or in mirrors.
- Causes driver to follow vehicles at shorter distances.
- Increases driver stress and frustration because of the additional mental workload.
- Reduces the driver's awareness of what is happening around them.

A range of things can make these outcomes worse, including the general situation on the road, how complex the use of the mobile phone is (e.g. texting versus answering a call), the duration of the task, and its frequency.

In addition, since mobile phone use is cognitive distraction, using it in the hands-free mode is similar to the risk associated with using it as a hand-held device¹⁰. For example, when drivers use the hand-held mode, they might slow down or maintain longer following distances. This would still be dangerous if the road situation is not suitable for slow driving. Also, when they use the hands-free mode, it is less likely that they do any of these "compensatory behaviours". Their speeds would stay the same since they probably do not associate any risk with using the phone hands-free.

While Bluetooth allows for hands-free calls so the driver's hands are only on the wheel and gear shift, using any device is still very distracting. Even with your mobile on Bluetooth or in a holder, drivers are often distracted by the phone conversation they are having or are trying to scroll through to find their favourite song to play next. Any time your eyes are off the road for more than one to two seconds, you are putting yourself, your passengers and those around your vehicle at risk. The mobile phone should be used only by other passengers OR in a medical emergency OR when staying in one place and pulled to the side of the road.

Check out these videos of overconfident drivers who think they can drive effectively while checking their mobile phone and what can happen when

⁹ Mobile phone use: A growing problem of driver distraction, https://www.who.int/violence_injury_prevention/publications/road_traffic/distracted_driving_en.pdf World Health Organisation, 2011.

¹⁰ Is a hands-free phone safer than a handheld phone? Ishigami, Yoko & Klein, Raymond, Journal of safety research, 2009.

distracted only for a few seconds. <https://www.autoblog.com/2018/09/11/teens-text-while-driving-distracted-study/> and <https://www.youtube.com/watch?v=Qel5LaudqO4>

The effect of text messaging on driving behaviour

Taking the specific act of texting and driving, the following types of distraction take place:

- Physical distraction through holding the phone
- Cognitive distraction through the mental effort used to focus on typing
- Visual distraction through creating and reading the messages
- Auditory distraction from hearing the sound of a notification

Reflection

1. What can distract you while using the road? List all the things that come to your mind and specify what type of distraction each is (visual, cognitive, physical, auditory).
2. What is the most common road behaviour you have seen on the road due to distraction?
3. What alternative do you have to the use of mobile phone while driving, walking or cycling?

Wrap-up

Distraction affects driver behaviour and increases the risk of a crash. There are many different kinds of distractions and their impact depends on their frequency and duration as well as the road situation. Drivers can avoid dangerous distractions such as mobile phone use by parking at the side of the road to respond to urgent calls or text messages, or by placing their phone out of reach while they are controlling a vehicle. There are also some applications (apps) that prevent the use of a phone while driving that can be downloaded to smart phones. Using these apps can be a useful way to resist the temptation to use a phone when attention should be on the road.

The fundamental principles of the international Red Cross and Red Crescent movement

Humanity The international Red Cross and Red Crescent movement, born of a desire to bring assistance without discrimination to the wounded on the battlefield, endeavours, in its international and national capacity, to prevent and alleviate human suffering wherever it may be found. Its purpose is to protect life and health and to ensure respect for the human being. It promotes mutual understanding, friendship, cooperation and lasting peace amongst all peoples.

Impartiality It makes no discrimination as to nationality, race, religious beliefs, class or political opinions. It endeavours to relieve the suffering of individuals, being guided solely by their needs, and to give priority to the most urgent cases of distress.

Neutrality In order to enjoy the confidence of all, the Movement may not take sides in hostilities or engage at any time in controversies of a political, racial, religious or ideological nature.

Independence The movement is independent. The National Societies, while auxiliaries in the humanitarian services of their governments and subject to the laws of their respective countries, must always maintain their autonomy so that they may be able at all times to act in accordance with the principles of the Movement.

Voluntary service It is a voluntary relief movement not prompted in any manner by desire for gain.

Unity There can be only one Red Cross or Red Crescent Society in any one country. It must be open to all. It must carry on its humanitarian work throughout its territory.

Universality The international Red Cross and Red Crescent movement, in which all societies have equal status and share equal responsibilities and duties in helping each other, is worldwide.

For more information on this IFRC publication, please contact:

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